IDDSI-Open v0: Code, Eval Results, and Smartphone Flow-Test Video Grader - Photo Classification Killed at Its Pre-registered Gates; Dataset and Weights Withheld

Technical note - v0 research preview (release shape locked 2026-09-07)

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Abstract IDDSI-Open v0 ships code plus eval results plus a deterministic computer-vision grader for the 10 mL / 10 s IDDSI Flow Test, plus a measured negative finding on photo classification. The model weights and the harvested image dataset are WITHHELD - a deliberate, stated decision. The dataset is withheld on licence grounds: of 587 unique events exactly 2 would ship cleanly (592 rows carry unverified open-images/openverse licences, 7 are CC BY-SA against the BY-NC-SA main release, 109 have unresolved redistribution status). The weights are withheld because the photo-classification head failed its pre-registered gates - wk1 weighted kappa >= 0.70 (measured 0.42), wk2 macro-F1 >= 0.75 (measured 0.358) with dangerous under-classification at 33% against a <= 5% gate - so the wk1 kill rule fired and photo-classification is dead as a product capability. The headline negative finding: two VLM judges (Qwen3-VL-2B vs Qwen3-VL-8B) agree on only **31.0%** of the same images (n_pairs = 575, full tier-2 sample, regen 2026-09-16; Cohen's kappa unweighted 0.1062, quadratic-weighted 0.2766 - still essentially chance for 5 classes), with a systematic directional bias toward under-predicting difficulty (Section 3.3) - because middle IDDSI levels (L4/L5/L6) need absolute size/flow information a scale-free web photo does not carry; every label-derived metric is therefore measured against noise. A CORAL ordinal-decode bug that made interior levels unreachable was found and fixed (held-out n = 109: macro-F1 0.088 -> 0.358, coverage 0.70 -> 0.88, ECE 0.138 -> 0.091, temperature 2.83 -> 0.93; weighted kappa flat at 0.42). The flow-test grader works: MAE 0.065 ml on a 100-video bench (L0-L4). Planned next step, not done: physically-tested desk specimens (60 base foods x 5 levels = 300, single-operator, protocol-following, test-filmed, with a filmed re-test subsample for intra-rater reliability). This is a research preview. Research demonstration for culinary education. Estimates visual similarity to IDDSI descriptors; does not perform official IDDSI tests, assess swallowing, determine suitability for any person, or decide whether food is safe to consume. The ambition was "the first publicly downloadable IDDSI-specific smartphone-food dataset and fine-tuned model released under explicit reuse licences, with physically tested Chinese/Cantonese soft-meal labels"; v0 does NOT meet it, and the ONE claim that stands is "the first open-source smartphone grader for the 10 ml IDDSI Flow Test". We claim neither clinical validity nor any ability to determine swallowing safety from a photograph.

Contents

1. Introduction 2
2. Related work 3
3. Methods 4
   3.1 Capture protocol and reference labels 4
   3.2 Data sources and provenance controls 5
   3.3 Level-compatibility model 5
   3.4 Deterministic Flow Test grader 6
   3.5 Evaluation plan 7
4. Generator failure case study 8
5. Regulatory posture 9
6. Limitations 9
7. Conclusion 10
References 10

1. Introduction

Dysphagia-safe food and liquid preparation depends on the IDDSI framework, whose levels are defined by physical tests (flow test, fork pressure, fork drip, spoon tilt), not by appearance. Existing automated approaches (Section 2) infer levels from images or video with accuracies ranging from 61.74% exact match (EdUHK, CEIV-115) to 96.5% controlled accuracy (ViscoCam), but none publishes a downloadable IDDSI-specific dataset plus fine-tuned weights under an explicit reuse licence, and none pairs smartphone media with physically tested labels for Chinese/Cantonese soft meals.

IDDSI-Open v0 contributes:

1. a frozen capture protocol and event schema pairing guided smartphone media (45° plate photo + 8-12 s fork-press / spoon-tilt / fork-drip clips; liquid side-view 10 mL / 10 s syringe flow videos) with contemporaneous physical tests performed independently by a registered dietitian (RD) and a speech-language pathologist (SLP), with third-person adjudication of disagreements - shipped as code and protocol, not as a downloadable dataset;
2. code and eval results for the photo-classification path (SigLIP-2 CORAL head with calibrated abstention) - the model weights are WITHHELD because the head failed its pre-registered gates, and the harvested dataset is WITHHELD on licence grounds; the result is a measured negative finding, not a product capability;
3. an open-source deterministic CV grader for the IDDSI Flow Test (10 mL / 10 s), claiming "first open-source smartphone grader for the 10 ml/10-second IDDSI Flow Test" - explicitly **not** "first video grader";
4. mandatory abstention: the system must answer "unclear - perform physical test" rather than guess.

Contribution claim (v0 truth; release shape locked 2026-09-07 - the honest release card `hf/model_README.md` is the authority):

> "The ambition was 'the first publicly downloadable IDDSI-specific smartphone-food dataset and fine-tuned model released under explicit reuse licences, with physically tested Chinese/Cantonese soft-meal labels'; v0 does NOT meet it - the harvested dataset is withheld on licence grounds (of 587 unique events exactly 2 would ship cleanly; 592 rows carry unverified open-images/openverse licences, 7 are CC BY-SA against the BY-NC-SA main release, 109 have unresolved redistribution status) and the model weights are withheld because the photo-classification head failed its pre-registered gates (wk1 weighted kappa >= 0.70 measured 0.42; wk2 macro-F1 >= 0.75 measured 0.358, with dangerous under-classification at 33% against a <= 5% gate; the wk1 kill rule fired and photo-classification is dead as a product capability). The ONE claim that stands is 'the first open-source smartphone grader for the 10 ml IDDSI Flow Test'."

We never claim: first IDDSI AI; first image classifier; first dataset/benchmark; clinically validated; diagnostic; medical-grade; or able to determine swallowing safety from a photograph (frozen never-claim list, r5_prior.md).

2. Related work

Five direct prior works frame the landscape (as stated in r5_prior.md):

- **ViscoCam (PolyU, 2021).** Smartphone video plus motion sensing to classify three IDDSI liquid levels - 96.5% controlled accuracy, but only >81% under extreme conditions (*ViscoCam* 2021). Exposes neither dataset nor weights.
- **OptiTexture (HKUST).** Combines RGB with a <$20 light-scattering sensor: 91.96% on 112 L3-L6 samples; its vision-only baseline was 69.64% (TODO TODO). Exposes neither dataset nor weights.
- **EdUHK (Frontiers in Nutrition, 2026).** Single-image Qwen pipeline achieved only 61.74% exact match on CEIV-115; open VLMs missed every liquid hazard ("TODO" 2026). CEIV publishes examples, not the full machine-readable corpus.
- **Automated photogrammetric syringe test (2019).** Automated video/photogrammetric measurement of the official syringe test appeared in 2019 (*TODO* 2019). (This is why we claim "first open-source smartphone grader for the 10 ml/10-second IDDSI Flow Test," not "first video grader.")
- **Safe Plate (Australia, proprietary).** Advertises photo-and-video IDDSI verification (*Safe Plate*, n.d.). Proprietary; no published dataset or weights.

Also relevant context from r5_prior.md: the official IDDSI iOS/Android app is a descriptor/test-reference library, not a classifier; Viscgo hardware, CF-200N texture analysers, and menu-software vendors (MealSuite/Dietech; Kewpie/UDF, Pulmuone, Hyundai Green Food, apetito/winVitalis, Hormel/Lyons, SimplyThick) classify or test known products rather than infer an unknown plate. No independent mainland-Chinese plate-photo classifier was found. Open FoodSense/PlateInsight resources concern generic perceived texture, not IDDSI.

3. Methods

This section specifies the frozen study design and the implementation as evaluated in v0. Classifier and flow-grader results are reported in Sections 3.3-3.5; v0 ships no downloadable dataset and no model weights.

3.1 Capture protocol and reference labels

The unit of observation is one independently prepared recipe batch or one freshly prepared liquid test. Stills, clips, retakes, and extracted frames from that preparation remain attached to one `event_id`; they are not treated as independent samples. Kitchen staff or interns capture media with a native phone camera, filters and portrait effects disabled where possible. The kit includes a standard metal fork, spoon, timer, and a non-food-contact reference card carrying a verified 15 mm feature and the event ID. Liquid events use the study-approved 10 mL syringe with a 61.5 mm calibrated barrel. The food and liquid procedures follow the current official physical-test instructions (IDDSI 2019).

Food events at declared L3-L7 include a whole-plate still from approximately 45 degrees and continuous 8-12 s guided clips. L3 and L4 require spoon-tilt and fork-drip views; L5 requires spoon-tilt and fork-press; L6 and L7 require fork-press. The clip begins before utensil contact and continues through release, drip, deformation, or rebound. Fork-press capture includes thumbnail blanching as a repeatable visual endpoint, not as an inferred force measurement. The entire utensil, sample, and scale reference remain visible. Mixed-component meals require an explicit study decision and are not silently assigned a level from a single component.

Liquid events at declared L0-L4 use one uninterrupted side-view Flow Test video. The full barrel, outlet, markings, and collection vessel are shown against a contrasting background; the syringe begins at exactly 10 mL at the recorded serving temperature. Recording begins before release, retains a visible or audible release cue, keeps the syringe vertical and the meniscus visible, and continues for at least one second after the 10 s reading. Wrong starting volume, uncertain release time, occlusion, camera movement, or an unreadable meniscus triggers a documented retake rather than an inferred result.

Each capture session records recipe, batch, kitchen, phone, operator, declared level, cuisine, serving temperature, lighting, plate, consent, and protocol-deviation metadata. Faces, patient identifiers, staff names, medication, order slips, identifying speech, and geolocation metadata are excluded. Recipe, batch, and kitchen groups are prospectively separated across `train`, `val`, and locked `external_test` splits.

For release-eligible real events, an RD and an SLP independently perform and record the applicable physical tests on the same batch at the same serving condition without seeing one another's result. A qualified third reviewer adjudicates disagreements. Only a final numeric adjudicated level enters the physically tested release; unresolved, invalid, pending, or protocol-deviating events remain quarantined or excluded. Tests were performed by LinguaLeap and participating assessors; they are not IDDSI certification.

3.2 Data sources and provenance controls

The canonical interchange is `events.jsonl` plus a `media/` tree, with one event per row and safe relative media paths. The schema records source, split, recipe/batch/kitchen groups, capture conditions, consent and privacy checks, media hashes and roles, declared/weak/user levels, independent physical-test levels, adjudication, and release eligibility. Four source classes are kept distinct:

1. **Synthetic (`source="synthetic"`).** Development-only photographs are prompted toward visible L3-L7 descriptors for pipeline development and representation-aware pretraining. The prompt target is stored only as `level_declared`; all physical-test fields are null. Rows may enter `train` or `val`, never `external_test`, and cannot be counted as physical ground truth, achieved coverage, validation evidence, or a product claim.
2. **Real-weak (`source="real-weak"`).** Licence-eligible third-party food images support pipeline development. Physical-test fields remain null; model-derived weak labels, when present, remain explicitly weak. These rows are restricted to `train` and retain a mandatory attribution ledger.
3. **Real-crowd (`source="real-crowd"`).** `level_user` records a contributor's statement, not a performed physical test. These events may support robustness or separately reported weak-supervision work, but they do not enter a physically tested benchmark. They remain release-ineligible while privacy screening is incomplete.
4. **Real-tested (`source="real"`).** Release eligibility requires the contemporaneous independent assessments and adjudication described in Section 3.1. Declared recipe levels, visual labels, synthetic targets, crowd statements, and model predictions are never substitutes.

Counts by source, level, and split are as recorded in the shipped manifests; the harvested image dataset itself is WITHHELD in v0 on licence grounds and no downloadable dataset ships. Release-licence audit (v0): of 587 unique events exactly 2 would ship cleanly - 592 rows carry unverified open-images/openverse licences, 7 are CC BY-SA against the BY-NC-SA main release, and 109 have unresolved redistribution status. Code ships under Apache-2.0; model weights are withheld on safety-gate grounds (Section 3.3). The project is not the official IDDSI website and is not IDDSI-endorsed.

3.3 Level-compatibility model

The primary still-image path uses `google/siglip2-base-patch16-224` as the encoder and a five-class ordinal head for L3-L7. The head implements monotone CORAL-style thresholds: a shared scalar projection and ordered learned cut-points are converted into a probability distribution over the five ordered levels. A `timm` ResNet-18 with the same ordinal head is the small-CNN baseline. The executable training configuration uses 224-pixel inputs, class-balanced CORAL loss, dropout, weight decay, mixed precision, and gradient clipping. It first trains the head with the encoder frozen, then unfreezes the final encoder blocks; the frozen configuration specifies five independent seeds. v0 ships the training and evaluation code with its recorded results only; no model weights ship.

Model selection, temperature calibration, and the confidence threshold are performed on validation data only. At inference, samples below the frozen threshold return `unclear - perform physical test`; an abstention is never silently converted to a level. Qwen3-VL-2B is reserved for grounded multilingual explanations and guided multi-frame reasoning. It receives the structured decision but cannot change it, and it is not the grading path. Explanation-adapter weights are withheld with the model weights.

A CORAL ordinal-decode bug made the interior levels (L4/L5/L6) unreachable: argmax over sigmoid-difference class masses could never fire for interior classes (proven numerically from the checkpoint). It was fixed with the ordinal rank rule `level = #{k : P(y>k) > 0.5}`. Held-out results (n = 109): macro-F1 0.088 -> 0.358, coverage 0.70 -> 0.88, ECE 0.138 -> 0.091, temperature 2.83 -> 0.93; weighted kappa flat at 0.42. The pre-registered gates all failed: wk1 weighted kappa >= 0.70 (measured 0.42 - the kill rule FIRES, so photo-classification is dead as a product capability); wk2 macro-F1 >= 0.75 (measured 0.358) with dangerous under-classification <= 5% (measured 33%). The ceiling finding (regen 2026-09-16 on n_pairs = 575, full tier-2 sample): two VLM judges (Qwen3-VL-2B vs Qwen3-VL-8B) agree on only **31.0%** of the same images (Cohen's kappa unweighted 0.1062, quadratic-weighted 0.2766) - still essentially chance for 5 classes - because middle IDDSI levels (L4/L5/L6) need absolute size/flow information a scale-free web photo does not carry. Every label-derived metric above is therefore measured against noise.

Under-prediction finding (new in v0.1, not in v0.0). The regen cross-tab on the full tier-2 sample (n = 575) reveals a systematic pattern the preliminary 22.6% figure did not state: tier-1 systematically *under-predicts* difficulty. Only L3 is reliably classified (11/12 correct). The interior classes drag toward the regular-food end - L4 calls (151) miss to L7 in 69 and to L6 in 28; L5 calls (131) miss to L7 in 54 and to L6 in 42; L6 calls (155) collapse into L7 in 111; L7 calls (126) over-pull to L6 in 47. This is the *dangerous direction* (predicting a SAFER texture than truth), the same direction the wk2 gate catches at 33% dangerous under-classification. Judges do not just disagree randomly; they share a directional bias toward the safe end. The remedy path is unchanged: physically-tested desk specimens.

3.4 Deterministic Flow Test grader

The video grader is a deterministic OpenCV pipeline with no learned model, network call, or GPU dependency. Its implementation-specific calibrated mode places a high-contrast ID-1 rectangle (85.60 x 53.98 mm) in the syringe plane to estimate pixels per millimetre and verify the 61.5 +/- 3.0 mm barrel. The grader detects paired barrel edges, checks verticality, rectifies the barrel, detects release from a nozzle change or the start of liquid motion, tracks the meniscus, and takes a robust reading at 10 s after release. Phone rotation metadata and non-ASCII input paths are handled during decoding.

Residual volume maps to L0 below 1 mL, L1 from 1 to below 4 mL, L2 from 4 to below 8 mL, L3 at 8 mL or above after visible flow, and L4 when no detectable flow occurs. A reading within 0.5 mL of the 1, 4, or 8 mL boundary abstains. The grader also abstains for a missing or unverifiable scale/syringe, excessive tilt, missing release, insufficient duration, low meniscus confidence, occlusion, or detected bubbles/lumps. It retains the numeric estimate and all quality checks for audit while

suppressing the level. Fiducial-optional mode is explicitly marked uncalibrated and is not used for the planned calibrated evaluation.

This contribution is described narrowly as the "first open-source smartphone grader for the 10 ml/10-second IDDSI Flow Test," not the first video grader, because automated photogrammetric measurement of the syringe test appeared in 2019 (*TODO* 2019). End-to-end performance (v0, the working deliverable): MAE 0.065 ml on a 100-video bench, grading L0-L4. A still photograph is never used to grade liquid flow.

3.5 Evaluation plan

Only `source="real"`, protocol-complete, adjudicated events in the locked `external_test` split contribute to the primary benchmark. Synthetic, real-weak, and real-crowd events are excluded or reported separately and are never cited as ground truth. The split is enforced at recipe, batch, and kitchen level. Calibration and abstention thresholds are frozen on validation data before external-test evaluation.

For the L3-L7 model, the v0 report contains macro-F1 with abstentions counted as misses; quadratic-weighted kappa on covered events; coverage and abstention rate; dangerous-direction under-classification (a prediction below the physical-test level); expected calibration error and temperature; and the CORAL-decode before/after comparison (Section 3.3). RD-versus-SLP raw agreement before adjudication would form the human inter-rater baseline but was not measured in v0; prespecified subgroup analyses and stratified bootstrap 95% confidence intervals are likewise not reported in v0.

The Flow Test evaluation compares estimated residual volume with the recorded physical reading, reports mL mean absolute error and L0-L4 agreement away from the predefined 1/4/8 mL boundaries, and separately reports boundary cases, abstentions, and technical failure reasons. Failure cases accompany held-out results. The project gates are prespecified in the runboard; failing a gate keeps the system at research-preview status rather than changing the metric or test set. In v0 the wk1 gate failed and the kill rule fired (Section 3.3): photo-classification is dead as a product capability, and only code, results, and the flow-test grader ship.

Observed metric (v0 / v0.1 regen)	Value
CORAL fix, held-out n = 109: macro-F1	0.088 -> 0.358
Coverage	0.70 -> 0.88
ECE	0.138 -> 0.091
Temperature	2.83 -> 0.93
Weighted kappa (covered events)	0.42 (flat before/after fix; measured against label noise)
Dangerous under-classification rate	33% (gate: <= 5% - FAILED)
wk1 gate (weighted kappa >= 0.70)	FAILED - kill rule fired
wk2 gate (macro-F1 >= 0.75)	FAILED (measured 0.358)
Inter-judge agreement (Qwen3-VL-2B vs Qwen3-VL-8B, n=575, v0.1 regen)	31.0% exact; kappa unweighted 0.1062, quadratic 0.2766 (the ceiling finding)
Under-prediction bias (v0.1 regen)	tier-1 systematically under-predicts L4-L6 toward L7; dangerous direction
Flow residual-volume MAE (100-video bench, L0-L4)	0.065 ml
RD-SLP inter-rater baseline	not measured in v0
Stratified-bootstrap 95% CIs	not reported in v0
Prespecified subgroups	not reported in v0

4. Generator failure case study

The first overnight synthetic-pipeline pilot on 1 September 2026 accepted **0 of 24** candidate images. This was a generator-quality-control outcome, not a model or dataset evaluation result. Manual audit agreed with the local Qwen3-VL-2B consistency judge and identified three recurring prompt-to-image failures:

1. **Grain texture ignored.** Images prompted as smooth L3/L4 masses retained visible whole rice grains or granular kernels.
2. **Garnish added.** The generator introduced sesame seeds, scallions, herbs, or other toppings whose untested texture contradicted the requested food presentation.
3. **Camera-facet nouns rendered as objects.** Terms intended to describe the capture style were interpreted literally, including a smartphone/device rendered in or on the food scene.

The prompt-bank remedy was applied before further GPU work: per-level visual guidance was strengthened with explicit smoothness, particle, garnish, and separated-liquid constraints; the common negative prompt now excludes phones/devices, whole or visible rice grains, garnish, and related artefacts; and camera-profile facets explicitly state that no device is visible in the scene. These constraints target all three audited failure modes while preserving capture-style variation.

No post-remedy acceptance rate or downstream benefit is claimed here; both are **TBD** pending a controlled rerun. The pilot reinforces the provenance rule that synthetic imagery is development-only and must be judged before use.

5. Regulatory posture

From r5_reg.md (frozen): as originally proposed (Snap-to-IDDSI), the system is probably Software as a Medical Device (SaMD), because it interprets an individual meal and recommends modification to mitigate a swallowing disorder; "Guidance only," "not a diagnosis," and "食品不能代替藥物" do not neutralize that intended medical purpose. Hong Kong: standalone medical-purpose software is SaMD under TR-007; MDACS listing generally voluntary. Mainland China: plausibly Class II, potentially Class III - obtain a formal classification determination before mainland release. EU: MDSW, Rule 11, Class IIa minimum, IIb arguable given choking/aspiration consequences. US: dysphagia-specific claims are not disease-unrelated "general wellness."

This technical note therefore describes an **open research/quality-assurance benchmark**, not a clinical or deployed safety classifier. Static RGB cannot directly measure flow, hardness, adhesiveness or cohesiveness; mandatory abstention and physical-test confirmation are essential (r5_prior.md, frozen verdict).

Frozen public-demo boundary wording (r5_reg.md):

> "Research demonstration for culinary education. Estimates visual similarity to IDDSI descriptors; does not perform official IDDSI tests, assess swallowing, determine suitability for any person, or decide whether food is safe to consume."

A regulated version would require fixed intended use, formal classification, ISO 13485/14971, IEC 62304/62366, locked/versioned model, analytical and clinical validation, subgroup/external-site evidence, cybersecurity/privacy, human-factors work, adverse-event/post-market processes, and jurisdictional submission (r5_reg.md). This note does not describe such a version.

6. Limitations

- **Photographs are not physical tests.** Static RGB cannot directly measure flow, hardness, adhesiveness, or cohesiveness (frozen verdict, r5_prior.md). A "not diagnosis" disclaimer does not override intended purpose (r5_reg.md).
- **Labels are not clinical ground truth for safety.** Tests were performed by LinguaLeap staff and participating assessors, not clinicians unaffiliated with the project, and they are not IDDSI certification (frozen wording, Section 3.1).
- **Weak/synthetic/crowd provenance classes** support pipeline development only; they must not be cited as ground truth (frozen, Section 3.2).
- **Synthetic generation is not self-validating.** The audited 0/24 pilot and its three failure modes are documented in Section 4. The prompt remedy has not yet been evaluated; its effect is TBD.

- **Photo classification killed at its gates.** The head failed every pre-registered gate (wk1 weighted kappa >= 0.70 measured 0.42 - kill rule fired; wk2 macro-F1 >= 0.75 measured 0.358, dangerous under-classification 33% against a <= 5% gate), and kappa 0.42 is measured against label noise: two VLM judges agree on only **31.0%** of the same images (n=575, kappa unweighted 0.1062, quadratic-weighted 0.2766 - still essentially chance for 5 classes), and tier-1 systematically under-predicts difficulty (Section 3.3). Weights are withheld; photo-classification is dead as a product capability.
- **Population and cuisine coverage:** unverified in v0. The planned remedy is physically-tested desk specimens (60 base foods x 5 levels = 300, single-operator, protocol-following, test-filmed, with a filmed re-test subsample for intra-rater reliability) - planned, not done.
- **Flow grader scope:** claims only the 10 mL / 10 s smartphone Flow Test grader (MAE 0.065 ml on the 100-video bench, L0-L4), not "first video grader" (2019 photogrammetric syringe measurement predates it).
- **What v0 reports and what it does not.** Classifier results (CORAL fix, gates, 31.0% ceiling finding + kappa + under-prediction) and the flow-grader bench are reported in Sections 3.3-3.5. The RD-SLP inter-rater baseline, subgroup analyses, and bootstrap CIs were not measured in v0. The 0/24 generator-QA pilot is not a model, dataset, or flow-grader evaluation.

7. Conclusion

IDDSI-Open v0 ships code, eval results, and an open-source smartphone grader for the 10 mL / 10 s IDDSI Flow Test (MAE 0.065 ml on the 100-video bench, L0-L4) - claiming only the narrow first stated in Section 1 ("the first open-source smartphone grader for the 10 ml IDDSI Flow Test") and never clinical validity - plus a measured negative finding: photo-classification failed its pre-registered gates (weighted kappa 0.42 against a >= 0.70 gate; macro-F1 0.358 against a >= 0.75 gate; dangerous under-classification 33% against a <= 5% gate) measured against **31.0%** inter-judge label noise on n=575 (kappa unweighted 0.1062, quadratic-weighted 0.2766, with a systematic tier-1 under-prediction bias - Section 3.3), so the model weights and the harvested dataset are withheld and photo-classification is dead as a product capability. Held-out results, failure cases, and test videos accompany the repository. Planned next step, not done: physically-tested desk specimens (60 base foods x 5 levels = 300, single-operator, protocol-following, test-filmed, with a filmed re-test subsample for intra-rater reliability). Numeric conclusions: as reported in Sections 3.3-3.5.

References

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Erratum and annotation notice · label set for every quoted kappa (H-YARDSTICK)

Added 2026-10-12 by lane rl03_yardstick_caveat. These pages are appended; every preceding page is unchanged, and no page number in this document's contents shifts because of them.

Every kappa quoted in this note was measured against a label set, and the gate outcome is a property of that label set rather than of the model (H-YARDSTICK, hf/HAZARD_LOG.md). The annotation below is carried inline in the source of this document (paper/paper.md and paper/paper.html) and in hf/model_README.md, hf/HAZARD_LOG.md and README.md. It is reproduced here because this PDF is a build artifact that could not be regenerated where the annotation was made: no TeX engine was available to run paper/build.sh.

The sentence this erratum adds to every kappa quoted above:

> The v0.4/v0.5 numbers were measured against the v0.5 consensus label set and fail the pre-registered wk1 >= 0.70 gate on it; the same models, measured against the v0.7 full-coverage consensus, pass it (v0.7: 0.838-0.860, 5 seeds, held-out n=108). Nothing about the models changed between those two columns - only the yardstick did.

Applied to the 17 kappa occurrences in this document, which sit on these pages:

> page 1 (abstract) 3 occurrences - the 0.42 model-eval figure and the inter-judge 0.1062 / 0.2766 figures
> page 3 (Section 1, contribution claim) 1 occurrence - the 0.42 model-eval figure
> page 6 (Section 3.3) 3 occurrences - the 0.42 model-eval figure and the inter-judge 0.1062 / 0.2766 figures
> page 7 (Section 3.5, evaluation plan) 1 occurrence - the metric the plan names; no value is quoted there
> page 8 (Section 3.5, results table) 3 occurrences - the 0.42 figure, the wk1 gate row and the inter-judge row
> page 10 (Sections 6 and 7) 6 occurrences - the 0.42 figure and the inter-judge figures in the limitations and the conclusion

Label set, stated per figure, because the two kinds of kappa are not comparable:

> - The 0.42 model-eval kappa (CORAL decode fix, held-out n=109) was measured against the Qwen tier-2 judge labels - the two-size, one-family weak label set of that run. It fails the wk1 >= 0.70 gate on that yardstick.
> - The v0.4/v0.5 model numbers were measured against the v0.5 consensus label set. They also fail wk1 >= 0.70 on it.
> - The 0.1062 / 0.2766 figures are agreement BETWEEN the two VLM judges who produced the label set. They characterise the yardstick itself rather than a model scored against one, so they have no model-kappa comparison of their own.
> - The same checkpoints measured against the v0.7 full-coverage consensus pass wk1: v0.4 0.736-0.808, v0.5 0.704-0.795, v0.7 0.838-0.860 (5 seeds each).

What this erratum does NOT do: it deletes no older number, moves neither the wk1 nor the wk2 gate, re-registers neither gate, does not change the release shape decided 2026-09-07, and does not alter the weight- and dataset-withholding decisions, which rest on the dangerous under-classification and calibration failures and hold on every yardstick. The wk2 33% failure is independent of the label set and stands on every reading.

Reproduce: paper/paper.md and paper/paper.html carry the same sentence inline at each of the sections listed above. Rebuild this PDF from paper.md with paper/build.sh once a TeX engine is available, and these pages become redundant.

Erratum and annotation notice 2 - metric, population, denominators (rel06_paper)

Added 2026-10-12 by lane rel06_paper. Appended after notice 1; every preceding page is unchanged and no page number shifts. paper/paper.md is authoritative; this PDF body is the build that predates the reconciliation, so it is corrected here rather than re-flowed.

1. EVERY KAPPA NOW NAMES ITS METRIC AND ITS POPULATION. Notice 1 above says the same checkpoints pass wk1 against the v0.7 full-coverage consensus. Stated in full:

   > wk1 is quadratic weighted kappa over COVERED ROWS. On that metric, against the v0.7 full-coverage consensus, held-out n = 108, all three checkpoints clear 0.70 in 5/5 seeds each - v0.4 0.736-0.808, v0.5 0.704-0.795, v0.7 0.838-0.860. On the ALL-ROWS kappa, which keeps abstentions in the denominator, the v0.5 model averages 0.671 and clears 0.70 on only 1 of 5 seeds (0.647-0.726). The pass verdict is therefore a property of the metric as well as of the yardstick (H-YARDEVID). `results/v07_release/bias_matrix.json` holds the all-rows quantity, not the gate metric; the covered-rows table is `results/yardstick_audit/per_seed.json`.

   The 0.4227 quoted throughout this PDF is quadratic weighted kappa over covered rows, held-out n = 109, 96 of 109 events covered, against the Qwen tier-2 judge labels. The inter-judge figures 0.1062 / 0.2766 are over all 575 judged pairs with no coverage filter, because neither judge abstains; they characterise the yardstick, not a model scored against one.

2. CORPUS DENOMINATORS. 152 events were never labelable. 152/684 = 22.2% of the whole pre-clean corpus is the rate to quote for any claim about the corpus (all 152 ids are present in the 684-event pre-clean manifests and none survives in the cleaned 532-row manifest, 684 - 152 = 532). 152/510 = 29.8% is a conditional rate over the both-judged subset and must not be quoted as a corpus rate: the aggregate is not self-consistent (683 + 512 - 510 = 685 against a corpus of 684). The v0.7 held-out test split is 108 of the 532 cleaned rows.

3. THE L5/L6 DECODE COLLAPSE, reported as a limitation. On the v0.7 held-out split (108 events, 5 seeds), per-level recall on decided events at seed 43 (92 decisions of 108) is L3 0.90, L4 0.50, L5 0.00, L6 0.00, L7 1.00, and L5 and L6 are never emitted as a decision in ANY of the five seeds (seeds 29, 71 and 101 emit no L4 either). The head is an L7 detector with a partial L3 detector, not a five-level classifier. Because the test split is 64.8% L7 (70 of 108), the always-L7 baseline scores 64.8% exact against the model's 80.9-84.9% on covered events and 66.7-70.4% on all 108 events with abstentions scored as misses.

What this notice does NOT do: it deletes no older number, moves neither the wk1 nor the wk2 gate, re-registers neither gate, does not change the release shape decided 2026-09-07, and does not alter the weight- or dataset-withholding decisions. No figure above changes sign; the direction of every prior finding survives on either metric alone.

Reproduce: `python3 -P tools/verify_paper_claims.py` recomputes every number in this notice from results/v07_release/v07_seed*_metrics.json, results/yardstick_audit/per_seed.json and results/v05_release/{labelability_sweep.json,unusable_event_ids.txt}, and lints paper/paper.md and paper/sections/*.md against the four rules above. Output: results/v07_release/paper_claims_verification.json. Tests: tools/tests/test_paper_claims.py.